IRON SUCROSE

Hypersensitivity — observe 30 min

Multiple small doses

CHEMICAL NAME Polynuclear iron(III)-hydroxide sucrose complex (20 mg elemental iron/mL)

CLASS Parenteral iron preparation

TRADE NAMES Venofer · Ferrosac · Iron sucrose injection

THERAPEUTIC USES / INDICATIONS

- Iron-deficiency anaemia when oral iron is ineffective, not tolerated or rapid repletion needed.
- CKD (especially haemodialysis, with ESA).
- Pregnancy (2nd/3rd trimester) with significant iron deficiency.
- Malabsorption, IBD.

ADMINISTRATION

- **IV** slow injection (undiluted, ≥2–5 min for 100–200 mg) or infusion diluted in saline only (e.g. 200 mg over ≥30 min).
- Observe for 30 min after each dose with resuscitation facilities available.
- Avoid extravasation (brown skin staining).

DOSAGES

- 100–200 mg IV per dose, up to 3 times weekly; max 200 mg per injection (500 mg infusion in some protocols).
- Total iron deficit by Ganzoni formula: weight (kg) × (target Hb − actual Hb, g/dL) × 2.4 + iron stores (500 mg).
- **Haemodialysis:** 100 mg per session × 10.

PHARMACOKINETICS

- Taken up by reticuloendothelial system; iron released to transferrin.
- **t½:** ~6 h.

PHARMACODYNAMICS

- Replenishes **iron stores** for erythropoiesis, faster than oral iron.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Taste disturbance (metallic).
- Headache, nausea.
- Hypotension, injection-site reactions.

ADVERSE EFFECTS (SERIOUS)

- Hypersensitivity/anaphylactoid reactions (rare but serious).
- Extravasation skin staining (can be permanent).
- Iron overload with excess dosing.

CONTRAINDICATIONS & CAUTIONS

- Anaemia not caused by iron deficiency; iron overload.
- Previous serious hypersensitivity to parenteral iron.
- First trimester of pregnancy.
- **Caution:** active infection/bacteraemia, asthma, atopy.

KEY INTERACTIONS

- **Oral iron:** ↓ absorption — stop oral iron during IV course (start 5 days after last injection).

NURSING MANAGEMENT

- Baseline BP, HR; monitor during and 30 min after; anaphylaxis kit ready.
- Check cannula; stop if pain/swelling (staining).
- Check Hb and ferritin ≥4 weeks after course (ferritin falsely high soon after).
- Document cumulative dose.

PATIENT EDUCATION

- Report itching, rash, breathlessness, dizziness or pain at the drip site.
- Do not take iron tablets during the IV course unless advised.
- A metallic taste can occur briefly.

KEY POINTS

- IV iron in repeated small doses (max 200 mg).
- Observe 30 min; hypersensitivity.
- CKD, oral-iron intolerance, pregnancy (2nd/3rd trimester).
- Extravasation causes staining.

SOURCES CHECKED **[929]** Venofer (iron sucrose) 20 mg iron / ml, solution for injection or concentrate for solution for infusion (UK SmPC) **[930]** Iron Sucrose, IV (US label) **[931]** Iron Sucrose, IV (US label, Mylan Institutional LLC)

Full references p. 432–457